Event based Experiential Computing in Agro-Advisory System for Rural Farmers

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***Abstract*—In our earlier proposed mKRISHI framework, which is a mobile phone based agriculture advisory system, a farmer can record a query with the help of audio-visual interface on the mobile phone application. The multi-modal sensory devices have been deployed to capture the context of the farm so that the query along with the context are presented to the agriculture expert at the web console for the investigation. In this paper, we propose an event based experiential computing approach for agro-advisory services. The classification and modeling of agricultural events, modeling of the agricultural experiences, and a method to browse through the history of agriculture experiences are the key contributions in the present work.**

I. Introduction

The mobile based agriculture advisory project, mKRISHI, [1], [2] was initiated with the goal of providing the benefits of information and communication technology (ICT) to rural farmers by enhancing their agricultural productivity, farming efficiency and improving their earnings. The goal of mKRISHI is to bridge the barrier between farmers and other stake-holders such as agricultural experts, agro-business units, financial institutions, hospitals and other utility providers, in their socio-economic ecosystem.

In this paper, we focus on interactions between rural farmers and agriculture experts from universities and research organizations. We develop approaches that allow a farmer to use audio-visual facilities that mKRISHI provides on a mobile phone to articulate their queries to experts with minimal use of text. An expert can only effectively advise a farmer if he has all information related to the farm available with him. The main contextual data elements of mKRISHI include many different types of sensors, such as temperature, humidity, soil moisture, canopy temperature, canopy humidity and wind velocity, placed on the field with data loggers to communicate the observations to the mKRISHI server. Apart from sensor information the farmer uploads information about climatic conditions, soil conditions, rain and fertilization history, and the pesticide and insecticide history. By presenting all this information in the context of the farmer query, the expert can diagnose the problem and promptly provide advice to the farmer in his native language and maybe even using audio-visual suggestions.

One of the most critical elements of mKRISHI's current and future functionality, and of many similar systems that one can easily envision, is how to present a large volume of spatial-temporal data from heterogeneous sensors in order to provide proper contextual information to the expert, without overloading him with voluminous details.

We are adopting *event-based experiential approaches* [3]–[5] to analyze data from different sources and develop techniques to store and visualize all this multi-modal data at the granularity relevant to the context. We are developing techniques to detect events at different sensor levels, combine different sensor-level events with other relevant events so as to detect meaningful agricultural events that are commonly used by experts. This paper presents our approach in this direction. We adopt Westermann-Jain model of events [4] to model event attributes at different levels and to represent composite events. By utilizing causality among events, we can help experts in analyzing situations. Further, the expert will be able to save and share the experience, diagnosis and prescription in specific events, for future use.

Our approach and experience with mKRISHI is a significant departure from existing systems for rural development, reviewed in Section II, in the use of multimedia at multiple levels and organizing information around events. We believe that the contribution in this paper is a first step in the direction of developing computing applications for rural masses.

II. Related research

This Section summarizes some of the recent advances in rural development, particularly in the field of precision agriculture.

A *Soil Test Crop Response Based Fertilizer Recommendation Software* [6] has been developed by Indian Institute of Soil Sciences, Bhopal, India. It currently provides fertilizer information to 11 states in India which is personalized based on district, soil type, crop, crop variety, season, target, and if available fertility status. In [7], the challenges involved in the development of decision support system to be used by farmers as end user are presented.

Sermed and Blackmore in [8], deals with the problem of optimizing irrigation for advantages such as conservation of water and energy, and improvement in the productivity. In [9], authors propose the use of Wireless Sensor Networks on the Australian agriculture fields to automate and optimize the irrigation to withstand the loss of water due to below-average rainfall. In Agrisens project [10], Wireless Sensor Network

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(WSN) had been deployed at Sula vineyard, Nasik, India, for precision agriculture. It addresses issues related to the productivity and disease prediction of the grape crop. It also implements the close looped system to automate and optimize the irrigation in the field. In [11], Burrell et al. performs the study of feasibility and impact of sensor network in agriculture by considering the example of a vineyard. The Lofar Agro Project [12] aims to monitor micro-climate in agricultural field using distributed computing in WSN. A decision support system (DSS) was proposed for providing insights to the applicability of pesticides, fertilizers, tillage and sowing rate. In [13], sensor network has been used to monitor climate, soil, pasture and animals and to form a closed loop control system.

In this paper, the mKRISHI project however aims to bridge the gap between farmers, agricultural experts, financial institutions, soil testing labs, agriculture market and other agriculture related institutions. We propose a novel experiential computing approach for the mKRISHI framework which aims to provide more insights to an expert by capturing, detecting, storing and analyzing the history of various events in agriculture.

III. Overview of mKRISHI System

In this Section, we present the architecture and goals of the mobile multimedia agro advisory system (mKRISHI) [1], [2].

A. Field Architecture of mKRISHI

The mKRISHI system consists of weather stations to collect sensor data, mobile phones with mKRISHI client application to be used by the farmers, expert's console accessible to agriculture experts and the mKRISHI server. Each weather station possesses atmospheric, soil and plant parameters monitoring sensors; data logger and modem for data storage and transmission; battery to energize all blocks of the weather station and a solar panel based battery charging unit. The sensors that are available with weather station includes temperature, relative humidity, soil moisture, soil temperature, grass temperature, wind direction, wind speed, solar radiation, rain gauge, leaf temperature and leaf wetness, and virtual dew point sensor. The data logger on weather station collects the data from sensors and transmits it to the mKRISHI server through a cellular network.

Each farmer, seeking the service, is initially required to perform registration by providing the details of the field location, crop, crop type, soil type, petiole analysis reports, and history of irrigation, fertilizer and pesticide application on the field. The *mKRISHI mobile phone application* allows the farmer to raise a query using text, voice, pictures and videos. The facility of sending multimedia information along with the query provides the appropriate context corresponding to the query.

The expert has access to all the content as well as contextual information related to the query of the farmer on the *expert's console*. The expert's console (Fig.1) shows the query attributes communicated by the farmer, that are farmer voice, text and the images associated with the query. In order to help the expert in decision making, the expert console also provides the information such as soil test report, petiole analysis report, history of the fertilizers applied, sensor data graphs, farmer's cell number, query date, pruning date, stage of the farming, variety and grafting.

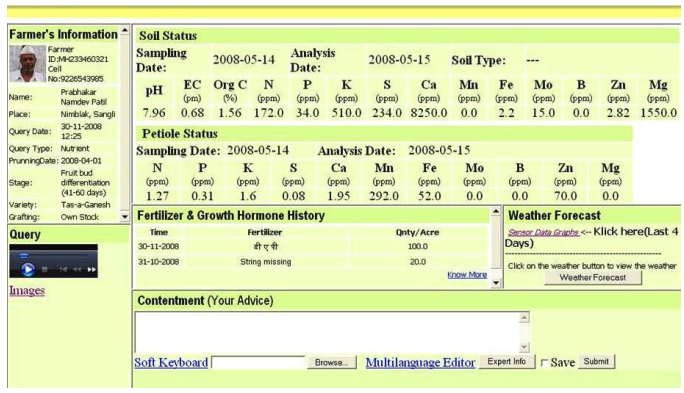

Fig. 1. A view of Expert's Console

The expert can respond to the farmer's query by referring to the above information on the expert's console, his experience and his agricultural knowledge. We provide a multi-language support on the console which helps the expert to reply in any language while typing in English. The expert can as well select one of the relevant responses from the list of stored responses available or upload a file as a response to the query. An option to save the responses for future referrals has also been made available.

In Fig. 2 a., the picture of infected cowpea crop as seen on the expert console is shown. While Fig 2 b. shows the response of the expert in the local language as seen on the farmer's mobile phone.

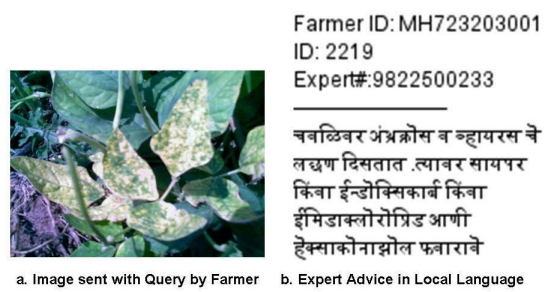

Fig. 2. a. Image of diseased (infected) leaves of cowpea crop sent along with voice query by a farmer.
b. Response in local language (Marathi) by an expert that is displayed on mobile phone of the user. This response means 'On cowpea, Anthracnose and Viral infections are suspected. Please spray Cyper or Indoxicarb or Imidacloroprid (pesticides) and Hexaconazole (fungicide)'.

B. Goals and Approach

The mKRISHI project aims to provide *expert advice* to the farmer's query with minimum delay, minimum service cost and maximum reliability. The main challenge is due to availability of few experts catering to the queries from a large number of farmers. The mKRISHI project also aims to alert

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the farmers in case of abnormal soil parameters and weather observations that lead to risks such as flooding, disease and pest attack.

In the subsequent sections, to motivate the experiential computing approach we present the learning from a case study in Section IV. The proposed *Event based experiential computing approach* is then presented in Section V. We model *agricultural experiences* which assists the expert to provide more reliable advice to the farmer's query. We also propose an efficient *cascading experience query system*, that browses through past relevant agricultural events with reduced delay. Further, *experiential computing* approach is used to generate the events, on occurrence of which, an alert message must be conveyed to the farmer.

IV. Learning from a case study

In this section, we present the learning from a case study by conducting survey in one of the villages in India. The data of yield outcome of 19 farmers of grape crop have been collected. Fig 3 shows the spread of the yield with respect to farming area. As expected, there is increasing trend of the yield with respect to increase in the farming area. However, the large variation in yield outcome for a particular farming area indicates the importance of the correct decision making in farming. Thus the best farming practice adopted by farmers producing more yield can also be adopted by the farmers producing less yield. This can be achieved through expert's advice based on the history of the experiences which motivates the proposed event based experiential computing approach for the agriculture advisory system.

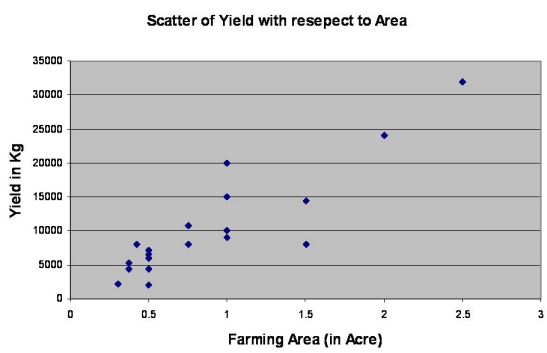

Fig. 3. Yield with respect to farming area for 19 farmers

V. Proposed Event based approach

Traditional computing has been considered to be document and data centric. Recently, in [3]–[5], [14], the concept of *Experiential computing* has been proposed to be a paradigm shift in computing. The concept is analogous to the information storage pattern of the human brain, which is in the form of events, rather than coarse data. The event based approach is very efficient in retrieval of past experiences which in turn helps in making crucial decisions, in the present scenario. The modeling of the event is proposed in [5], in which the event is represented by its various properties like time, location, type, informational and experimental attributes. A similar approach has been adopted in [15], in which the problem of identification of *context* using sensors, such as microphone, camera, accelerometer, temperature, humidity and light, embedded on the mobile devices, has been considered.

Based on the Westermann-Jain Event model [4], we propose the *experiential computing model* for the mKRISHI application. Here, the key idea is to identify the events that have significance to agriculture. The information from various sources (sensors, farmer, news, etc.) is stored in the form of events in an *Event-Base*. Further these events are linked in terms of various attributes (spatial, temporal, structural, causal and informational) which makes the retrieval of appropriate events easier. If each event is considered as a node, the causal linkages between various events will result in an *Event-Web*. While browsing through this Event-Web, one can seek insights and past experiences which helps in improving the present decisions. This approach helps to provide the history of agricultural experiences to the agricultural experts, which will in turn improve their responses to the farmer's query. Using this experiential model, less experienced experts can also make more reliable decisions. This solves the critical problem of mapping small number of experts to large number of queries.

In the rest of this Section, we first present the classification and modeling of events related to the mKRISHI application, followed by the modeling and browsing of experiences in mKRISHI.

A. Event Classification in mKRISHI

For classifying the events related to mKRISHI, we propose an extensible tree based structure (Fig. 4) where leaves of the tree represents the agricultural events. In mKRISHI architecture, the events related to mKRISHI are defined and classified in the following main categories.

1) Sensor based Event: As an example, a simple *binary* event may occur when any parameter observed by a particular sensor (say soil moisture sensor) is abnormally *High* or *Low*. The sensed data of any sensor when found above and below some predefined upper and lower thresholds, respectively represent the two event hypotheses, $\{High, Low\}$. Let, $X_{S.S.M}$ be the value observed by soil moisture sensor, then the binary *Sensor.Soil.Moisture* event $E_{S.S.M}$ is defined as

$$\begin{aligned} X_{S.S.M} &> T^{H}_{S.S.M} && ; E_{S.S.M} = High \\ X_{S.S.M} &< T^{L}_{S.S.M} && ; E_{S.S.M} = Low \end{aligned}$$

where, $T^{H}_{S.S.M}$ and $T^{L}_{S.S.M}$ are the predefined upper and lower thresholds for detecting the soil moisture event. Events can be detected using different sensors available in the present mKRISHI application as depicted in Fig.4.

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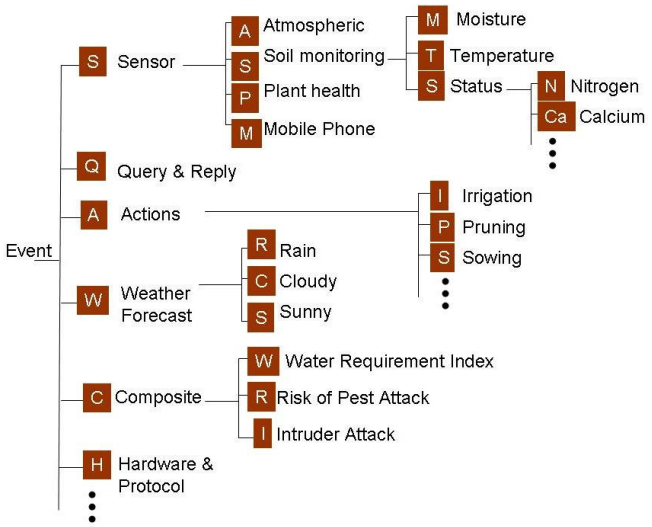

Fig. 4. mKRISHI Events Classification

2) Farmer's query and Expert's Advice: Using the mKRISHI client application on the cell phone, a farmer can raise the query using text, voice, pictures and video. This is further made available on the expert's console for an expert's advice. Any *query* raised by a farmer is considered as an *event*. Similarly, the *response* of the Expert to the query raised by the farmer can be considered as another *event*. This event gives the option to an expert to revisit the past similar queries of the farmers; the responses provided to those queries by the experts; the measures taken (actions performed by farmer) and the resultant consequences.

3) Actions performed by farmers and Stages in Farming: The *actions* of the farmers like spraying of pesticides, application of fertilizers and irrigation are considered as events. Further, various *stages* in farming like pruning, sowing, dormant, bud burst, various levels of shoots, various levels of blooms, berry touch, green fruit, variation, colored fruit and harvesting are also considered as events. The information about this category of events can mostly be obtained from the particular farmer itself. In the present mKRISHI system, a user-friendly application on the farmer's cell-phone allows updating of these types of events.

4) Weather Forecasts: The *weather predictions* in a certain region forms another category of events. For example, forecast of excess rainfall may be an important event to be detected, since it helps to take decisions on irrigation, pruning, and applications of pesticides. Thus when informed about the weather forecast events, the farmer may initiate some activity which in turn triggers another set of events. The strong linkages of the *weather forecasting events* with the expert's decision and farmer's action, makes it more important.

5) Composite Events: Although individual sensor readings based events are useful for farmers to make decisions, most important events are detected by combining observations made by many heterogeneous sensors. The problem of *multi-modal data fusion* to detect the composite events is thus important.

Plant disease risk is an example of a composite event. In agriculture literature (as summarized by Madden et al. in [16]), there exists various crop-specific models to predict the risk of the disease or pest attack on the crop. In [2], we present late blight disease forecasting protocol, by integrating sensor-based mathematical disease forecasting models, with human participatory diagnosis using mobile phone application overlay mKRISHI system. These models require parameters collected using soil, plant, and weather sensors. The risk index obtained will guide the farmer to take measures to avoid or control the disease.

6) System Hardware and Protocol: In our operational experience of mKRISHI, many events related to the hardware and protocol of the system have been observed. Some of them are - addition of new feature like SMS alert of rainfall forecast, upgradation of sensors, change of protocol version etc. These kind of events may also have some impact on the agro-advisory system and hence they are worth considering.

B. EVENT MODELING in mKRISHI

In Event-Web terminology [4], [5], each *event* is considered as an *event node*. Different such event nodes can be interconnected to form the web of events, called *Event-Web*. Each event node is associated with six attributes, that are: type, spatial, temporal, informative, experiential, causal and structural attributes. After an event is detected, the attributes of the events are identified and stored in the database of events called Event-Base. We extend the event modeling definition in [4], to events in the mKRISHI system. In the subsequent part of the section we define these six attributes of event in the agriculture context.

1) Type of Event attribute: The *type* of the event is a basic attribute to categorize various events stored in an event-base. Thus, a query for a particular event in an event-base can be made using the type attribute of that event. The event classification done in the previous sub-section can be used to assign an identification code associated with each event (based on its type) in an event-base.

2) Spatial attribute: The location of the occurrence of any event identifies the place where an appropriate action might be required. Currently in mKRISHI, most of the sensors are placed on the weather station whose (latitude-longitude) position is obtained from Global Positioning System (GPS). Thus in mKRISHI the location of the weather station will provide the spatial co-ordinates of the event. On the other hand, if the event like discoloration of leaves is observed by a farmer and the image is taken using the mobile phone for querying, then the GPS on the mobile phone, or the associated base station, will provide the latitude-longitude information of that event. While, if the event is a query by a farmer, then the location of the farm registered with the cell phone or the plot number provided by the farmer while asking the query will represent the spatial attribute of the event. For composite events, the location identified by the model will determine the spatial attribute of the event.

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3) Temporal attribute: Time of the occurrence of an event provides important historical information. Different events at a given spatial location may be linked with each other in temporal dimension. For example, excessive rainfall over a field may lead to an excessive soil moisture level which in turn results in an increased risk of diseases on the crop. These three events are interlinked in time with some delay in between the two events. The time stamp associated with any event may be captured using the time stamp along with the observation made by the sensors. Thus while logging the data, we have to have time stamp associated with the observation of the sensors, event captured by camera of the cell phones, external agencies forecasting rain and also with the query and various stages in the farming as updated by the farmer.

4) Informational attribute: On many occasions, along with the information about event occurrence, the raw information (content) is also important. For example, if the event *high rainfall* is detected, then the actual rainfall measurement should also be provided for reference. The actual sensor reading, actual query of the farmer, the details about current stage of the farming, the details of the weather forecasting, the actual index of the water required and the disease risk index are recommended to be stored in the event-base representing the informational attributes of the corresponding event.

5) Experiential attribute: The context in which an event occurs has been considered by [3] as an experiential attribute of that event. Korpipää et al. in [15], discusses identification of different context from various sensors on the mobile phone. This additional information, if available, which captures the context of the event along with its content (Informational attribute) can be stored in the event-base. For instance, if the event is assumed to be *spots on the leaves*, then the information will be the picture of the leaves, and the experiential attribute will be the farmer's insight which is captured using a microphone sensor on the cell phone. However, if the event is assumed to be the *farmer's query*, then the information becomes the actual query asked, while picture of the leaves will becomes an experiential attribute.

6) Causal and Structural attribute: These attributes define the inter-linkage between the different event-nodes hence forming an Event-Web. The causal information of the linkage between any two events along with the structure of the linkage is captured in the Event-Base.

C. EXPERIENCE MODELING and BROWSING

For each query it becomes difficult to analyze the history of vast data, which has been made available at the expert's console. This is because, the extraction of relevant information of a similar type of query raised along with its consequences in the past becomes difficult. Here, we propose to assist the expert with a more powerful *event based experience browsing* scheme.

1) Experience Modeling: After classification and modeling of the events in mKRISHI, the linkages between the events will provide the experience sharing platform between different experts.

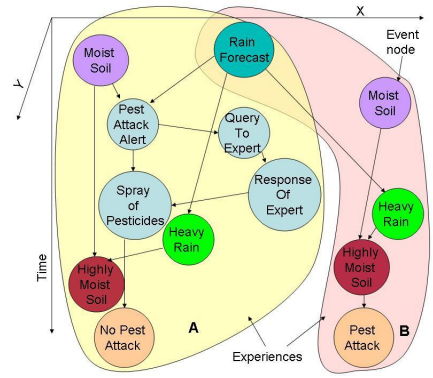

Fig. 5. An example of an experience obtained by linking of various events in spatial and temporal dimensions. Here, each node is a different event that has occurred at a different spatial-temporal coordinate.

In Fig. 5, we present an example of interlinked events to form experiences. Each event is shown as a node occurring at a particular time and 2-dimension space coordinates $(X - Y)$. The nodes are shown to be linked by causal linkages. Cluster of the the linked events creates an experience. Fig. 5 shows event nodes like: forecasting of rain, moist soil, alert to the farmers regarding pest attack, query by a farmer regarding type of pesticides to be used, response by an expert, spray of pesticides, occurrence of heavy rain, highly moist soil, occurrence or avoidance of pest attack; along with their linkages. Fig. 5 shows two clusters of linked events resulting in the following agricultural experience.

A. Due to proper expert's advice and timely application of pesticides the pest attack on the crop was successfully avoided.

B. Here similar kind of events resulted in a pest attack because pesticide was not applied when it was required.

2) Experience browsing: Here, we present a novel *cascading spatial-temporal query scheme* to efficiently browse the history of farming experiences. According to the event classification, let us assume the occurrence of the simple event of the soil moisture sensor observing abnormally high moisture levels at some location. Among the two outcomes $\{High, Low\}$, the event occurred is $E_{S.S.M} = High$. Meanwhile there is a query by a farmer related to the possible disease risk, that forms another event. Without loss of generality, for the present example the proposed cascading spatial-temporal query scheme is illustrated in the following two steps.

Search level 1: The events similar to the one which has occurred (say $E_{S.S.M} = High$) at the present instant, are searched in a limited nearby space and in a limited past time window. Fig. 6 shows a cube in spatial-temporal dimension in which the search for event of interest is required to be performed. This search may or may not result in finding the required related events. In case, it results in some similar events in the prescribed search volume, further search is performed as mentioned in Search level 2. Otherwise, the scope (spatial-temporal window) of search level 1 is broadened.

Search level 2: Now another event of concern (the query on disease risk asked by farmer) should be searched only in

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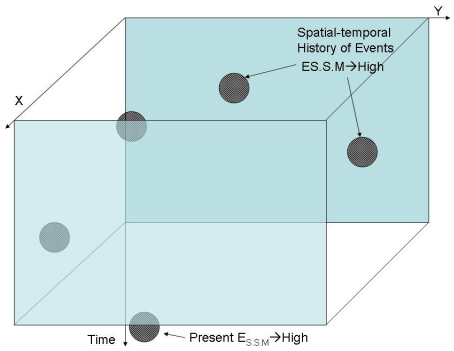

Fig. 6. Search level 1

near proximity (in time and space) of the resultant events in the previous search (see Fig.7). Without loss of generality, a

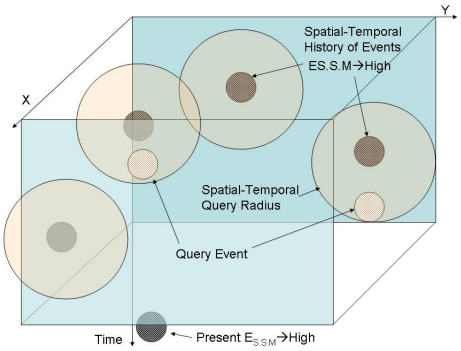

Fig. 7. Search level 2

sphere is shown surrounding the earlier identified events from Search level 1, in which the search for another event is carried out. However, the shape of the region in which the search is performed need not be a sphere. Further, the advantage of searching the event only in the near proximity and not in the whole Event-base is to conserve browsing time and cost. Considering the same example, if the event of concern is high risk of disease then a search should be performed for this event near the earlier detected history of High soil moisture event in Search level 1. This will indicate the probability of occurrence of the resultant event when the causing event (direct or indirect) had already occurred. As seen in Fig. 7, the explicit direct linkage between the two events is not shown. This is to reflect the fact that even if the events are indirectly linked, the chances of occurrence of the event in query may not be ruled out.

VI. Conclusions

A novel approach of Event and Experience based modeling in a mobile multimedia agro-advisory system had been proposed in the paper. Typical agricultural events were defined and a classification of these events was provided. The proposed event model was then used to build an event-base. The formation of agricultural experience was illustrated by examples. An efficient event querying model was proposed which helps to query the history of events and their linkages in spatial-temporal dimensions. The expert can browse through this history of linked events while responding to the query of the farmer. The proposed model provides a novel **experience sharing platform** for the mKRISHI agro-advisory system.

VII. Acknowledgments

We acknowledge other team members at TCS Innovation Labs, Mumbai, for their support in conceptualization and development of the mKRISHI system.

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